THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS X — SCIENCE (086) · FULL SYLLABUS PRACTICE PAPER

Session 2026–27 | CBSE / RBSE Pattern | Section A Biology · Section B Chemistry · Section C Physics

Class: 10	Subject: Science	Set 03	Time: 3 Hours	Max. Marks: 80

General Instructions:

1. This question paper consists of 39 questions in 3 sections. Section A is Biology, Section B is Chemistry and Section C is Physics.
2. All sections are compulsory. However, internal choice is provided in some questions. A student is expected to attempt only one of these questions.
3. Draw neat, labelled diagrams wherever required. Use of calculator is not allowed.

SECTION A — BIOLOGY — 16 questions · 30 marks

Q1–Q7 MCQs. Q8–Q9: Assertion (A) / Reason (R) – choose A, B, C or D.

1. A destarched potted plant is placed in a bell jar along with a dish containing potassium hydroxide solution and kept in sunlight for some hours. A leaf from the plant is then tested with iodine solution and does not turn blue-black. This shows that: **1**

A. light is necessary for photosynthesis
B. chlorophyll is necessary for photosynthesis
C. carbon dioxide is necessary for photosynthesis
D. oxygen is necessary for photosynthesis

2. A patient has had a large part of his small intestine surgically removed. Which of the following functions of his digestive system will be affected most? **1**

A. Absorption of digested food
B. Churning of food
C. Digestion of proteins by pepsin
D. Secretion of saliva

3. Ravi, a 15-year-old boy, is much shorter than other boys of his age though he is otherwise healthy. A doctor suspects a deficiency of a hormone secreted by: **1**

A. thyroid gland
B. adrenal gland
C. pituitary gland
D. pancreas

4. The correct sequence in which an impulse travels through a single neuron is: **1**

A. axon → cell body → dendrite → nerve ending
B. dendrite → cell body → axon → nerve ending
C. cell body → dendrite → nerve ending → axon
D. nerve ending → axon → cell body → dendrite

5. In human males the testes lie outside the abdominal cavity in the scrotum because: **1**

A. sperm formation requires a temperature lower than the normal body temperature
B. they need more space to grow
C. they are secretion glands which need air
D. they are protected by the scrotum from infection

6. Which of the following traits can be inherited by a child from its parents? **1**

A. Sun-tanned skin
B. Strong muscles developed by exercise
C. Eye colour
D. A scar on the face

7. In the food chain grass → insect → frog → snake → eagle, the secondary consumer is: **1**

A. insect B. frog C. snake D. eagle

8. **Assertion (A):** Reflex actions are very quick responses to stimuli. **1**
Reason (R): In a reflex action the impulse is processed in the spinal cord and does not have to travel to the brain and back.

A. Both A and R are true, and R is the correct explanation of A.

B. Both A and R are true, but R is not the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

9. **Assertion (A):** The flow of energy in a food chain is unidirectional. **1**
Reason (R): The energy captured by autotrophs does not revert to the sun and is lost as heat at every trophic level.

A. Both A and R are true, and R is the correct explanation of A.

B. Both A and R are true, but R is not the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

10. Name two waste products that plants eliminate. By what methods do they get rid of them? **2**

11. Name the part of the female reproductive system in which (i) fertilisation takes place and (ii) the embryo is implanted. **2**

OR

Write two functions of the testes in a human male.

12. A pure tall pea plant is crossed with a pure dwarf pea plant and all the F_1 plants are tall. Which trait is dominant? Write the genotype of the F_1 plants. **2**

13. Explain the mechanism of inhalation in human beings with reference to the diaphragm and the rib cage. How does the air pressure change inside the chest during inhalation? **3**

14. What are trophic levels? Why is the number of trophic levels in a food chain generally limited to four or five? **3**

15. In Rohan's family, free earlobes (F) are dominant over attached earlobes (f). Rohan's father has free earlobes, his mother has attached earlobes, and Rohan himself has attached earlobes.

(a) What is the genotype of Rohan's mother? 1

(b) What is the genotype of Rohan's father? Give a reason. 1

(c) Draw a Punnett square for the cross between Rohan's parents and find the chance that their next child has attached earlobes. 2

OR

(d) Explain with an example how a trait may be inherited but may not be expressed in an individual.

16. (i) Draw a labelled diagram of a neuron. (ii) Describe how an impulse is transmitted from one neuron to another at a synapse. (iii) State one function each of the cerebrum, cerebellum and medulla. **5**

OR

(i) Describe binary fission, budding and fragmentation with one example of each. (ii) What is regeneration? Name an organism that shows it.

SECTION B — CHEMISTRY — 12 questions · 25 marks

Q17–Q21 MCQs. Q22: Assertion (A) / Reason (R) – choose A, B, C or D.

17. In the reaction $2CuO + C \rightarrow 2Cu + CO_2$, the substance that is oxidised is: **1**

A. CuO B. C C. Cu D. CO_2

18. White silver chloride kept in a china dish turns grey when left in sunlight. This change represents: **1**

A. a photolytic decomposition reaction
B. a thermal decomposition reaction
C. a displacement reaction
D. a neutralisation reaction

19. The number of molecules of water of crystallisation in washing soda is: **1**

A. 2 B. 5 C. 7 D. 10

20. A zinc ore of the type $ZnCO_3$ is first converted to zinc oxide by a process called: **1**

A. roasting
B. calcination
C. smelting
D. froth flotation

21. The alcohols CH_3OH, C_2H_5OH and C_3H_7OH form a homologous series. The next member of the series is: **1**

A. C_3H_8O B. C_4H_9OH C. C_4H_{10} D. $C_5H_{11}OH$

22. **Assertion (A):** Ionic compounds have high melting and boiling points. **1**
Reason (R): There are strong electrostatic forces of attraction between the oppositely charged ions, and a large amount of energy is required to break them.

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, but R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

23. When green crystals of ferrous sulphate are heated, the colour changes and gases with a characteristic smell are evolved. State the colour change and write the chemical equation. **2**

24. A person suffering from acidity is given baking soda. Write the chemical equation for the action of baking soda on stomach acid and state why it is a safe antacid. **2**

25. (a) Why do acids conduct electricity in aqueous solution but not in dry form? (b) How would you test the gas evolved when an acid reacts with a metal carbonate? Write the equation involved for the test. (c) What is the role of water in showing acidic character? **3**

OR

How is washing soda obtained from sodium chloride? Write the chemical equations involved.

26. Write the name and the structural formula of: (a) the alcohol having three carbon atoms, (b) the carboxylic acid having two carbon atoms, (c) the ketone having three carbon atoms. **3**

27. Manish noticed that a silver spoon kept in the kitchen has turned black and a copper vessel kept in a humid place has become coated with a green layer, whereas gold ornaments remain shiny. His teacher explained that these changes are examples of corrosion.

(a) Name the black coating on silver and the green coating on copper. 1

(b) Why does gold not corrode? 1

(c) Name the two conditions necessary for the rusting of iron. Give one method to prevent it. 2

OR

(d) Why is aluminium not corroded easily although it is more reactive than iron?

28. (a) Explain the cleansing action of soaps with the help of a diagram of a micelle. (b) Why do soaps not form lather with hard water? (c) State two advantages of detergents over soaps. **5**

OR

(a) What is rancidity? State two ways of preventing it. (b) What is corrosion? Write the equation for the rusting of iron in words. (c) Balance: $Fe + H_2O \rightarrow Fe_3O_4 + H_2$.

SECTION C — PHYSICS — 11 questions · 25 marks

Q29–Q32 MCQs. Q33: Assertion (A) / Reason (R) – choose A, B, C or D.

29. A truck driver uses a rear-view mirror that always gives an erect, diminished image of the traffic behind. The mirror is: **1**

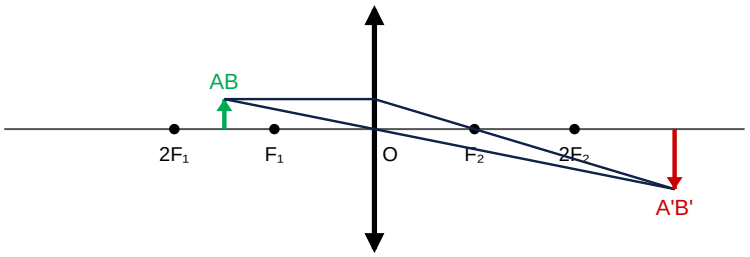
A. convex

B. concave

C. plane

D. a concave mirror used with the object beyond C

30. A convex lens of focal length 10 cm forms an image of an object placed 15 cm in front of it, as shown. The image is formed at a distance of: **1**



A. 15 cm, real and same size

B. 30 cm, real, inverted and magnified

C. 30 cm, virtual and erect

D. 7.5 cm, real and diminished

31. Danger signals are red in colour because red light: **1**

A. is scattered the most by smoke and fog

B. has the shortest wavelength

C. is scattered the least by smoke and fog and so can be seen from far

D. has the maximum speed in air

32. The SI unit of resistivity of a material is: **1**

A. ohm B. ohm per metre C. ohm metre D. siemens

33. **Assertion (A):** Magnetic field lines never intersect each other. **1**
Reason (R): If two field lines intersected, there would be two directions of the magnetic field at the point of intersection, which is not possible.

A. Both A and R are true, and R is the correct explanation of A.

B. Both A and R are true, but R is not the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

34. What is meant by the near point of a normal human eye? Give its value. **2**

OR

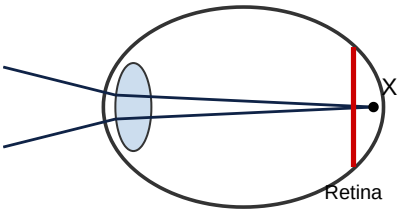
Why do we see the Sun about two minutes before the actual sunrise?

35. An object 4 cm high is placed 20 cm in front of a convex mirror of focal length 20 cm. Find the position, nature and size of the image. **3**

36. An electric heater of resistance 44 Ω is connected to a 220 V supply. Calculate (a) the current drawn, (b) the power of the heater and (c) the heat produced in 10 minutes. **3**

37. Describe an activity to show that a current-carrying conductor placed in a magnetic field experiences a force. How can the direction of the force be reversed? **3**

38. Ria's grandfather finds that he can see the newspaper clearly only when it is held at arm's length (about 50 cm). He cannot read it comfortably at the normal reading distance of 25 cm, though his distant vision is perfect. His doctor prescribes spectacles.



Parallel rays would meet at X, behind the retina

(a) Name the defect of vision from which he suffers. 1

(b) State one cause of this defect. 1

(c) Find the power of the lens he should use to read at 25 cm. 2

OR

(d) What is a bifocal lens? Why is it prescribed for some elderly people?

39. (a) Derive the formula for the equivalent resistance of two resistors R_1 and R_2 connected in parallel. (b) Two resistors of 6 Ω and 12 Ω are connected in parallel across a 12 V battery. Find the current through each resistor and the total current. **5**

OR

A 3 cm tall object is placed 12 cm in front of a concave mirror of focal length 8 cm. (a) Find the position and size of the image. (b) Draw the ray diagram. (c) State the nature of the image.